Quantum Computation and Information

# Homework Set 4

1. **[Phase-flip codes]** Recall the phase-flip error $\{ \begin{smallmatrix} |0\rangle \to |0\rangle \\ |1\rangle \to -|1\rangle \end{smallmatrix}$ , and the states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The following $[3, 1]$ codes can be used to detect and correct the phase-flip error:

$$\{ \begin{smallmatrix} |0\rangle \to |\bar{0}\rangle = |+++\rangle \\ |1\rangle \to |\bar{1}\rangle = |---\rangle \end{smallmatrix} .$$

   (a). Draw the quantum circuit for encoding the above phase-flip codes.

   (b). Give the stabilizers that can be used to detect the phase-flip error.

   (c). Continuing on the encoding circuit from (a), draw the circuit for error detection and correction.

2. Recall Pauli X, Y, Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

   (a). Show the following anti-commutation relationships:

$$\{\sigma_i,\ \sigma_j\} = 2\delta_{ij} I,$$

   where $\delta_{ij}$ is the Kronecker delta.

   (b). Show the following commutation relationships:

$$[\sigma_j,\ \sigma_k] = 2i\epsilon_{jkl}\sigma_l,$$

   where $\epsilon_{ijk}$ is the Levi-Civita symbol.